A Case Study Report on

"TITANIC SURVIVAL PREDICTION USING LOGISTIC REGRESSION"

for the course

Fundamentals Of Data Science (22CSC59)

Submitted in partial fulfillment of the requirements for the award of degree

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

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1. PROBLEM STATEMENT

The Titanic Survival Prediction case study applies supervised machine learning to estimate whether a passenger survived the Titanic disaster. The dataset contains historical passenger records, including information such as passenger class, sex, age, number of siblings or spouses aboard, number of parents or children aboard, ticket fare, and port of embarkation.

The target variable is Survived. A value of 1 represents a passenger who survived, while a value of 0 represents a passenger who did not survive. Since the target has two possible classes, the task is a binary classification problem.

The main objective is to examine the dataset, identify useful patterns, prepare the data for modelling, train a Logistic Regression classifier, and evaluate its predictions using multiple metrics. The study also interprets the model's results and discusses limitations. The aim is not to prove why a passenger survived, but to identify patterns in the available historical data and assess how well a model can classify the outcomes.

1.1 Objectives

- Inspect the dataset structure, feature types, summary statistics, and missing values.
- Use visualizations to examine survival distribution and relationships between passenger characteristics and survival.
- Handle missing numerical and categorical values and encode categorical features.
- Train a Logistic Regression model using a reproducible train-test split.
- Evaluate the model using accuracy, precision, recall, F1-score, ROC-AUC, a classification report, and a confusion matrix.
- Interpret the findings and state the limitations of the analysis.

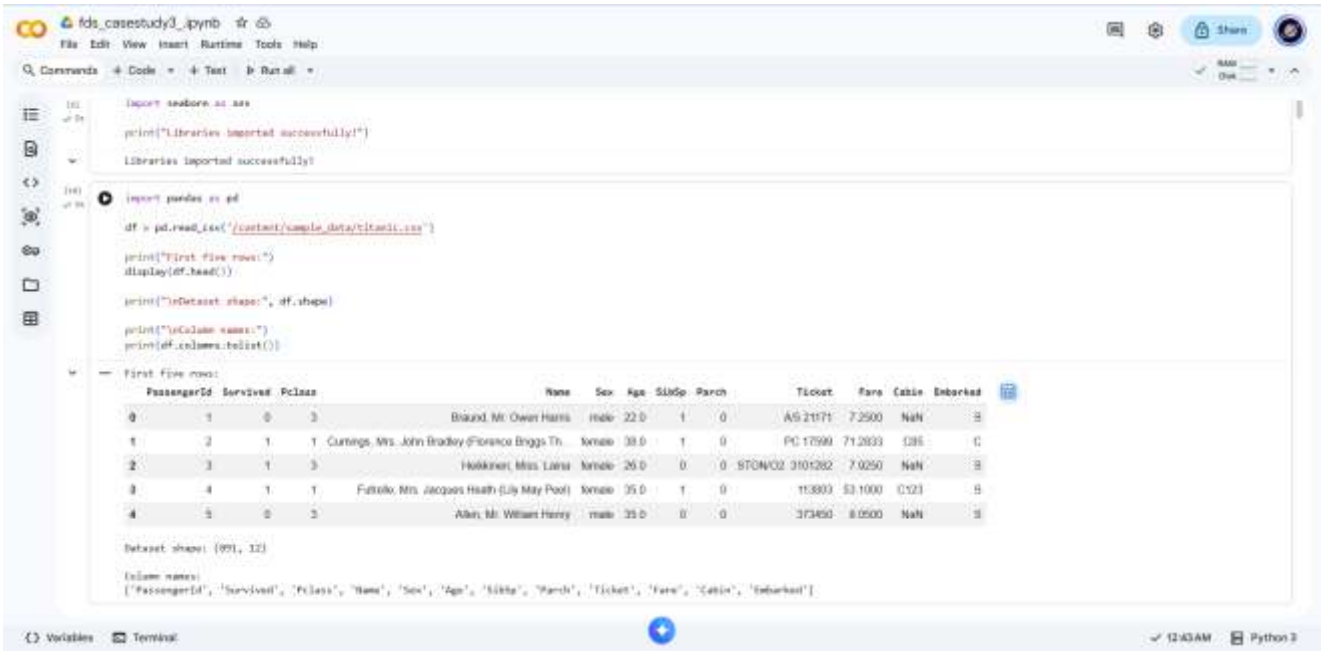
2. METHODOLOGY

The workflow begins with loading the Titanic CSV file into a Pandas DataFrame. The data is inspected and explored before model training. A preprocessing pipeline is then used to handle missing values, scale numerical features, and encode categorical features. Logistic Regression is trained on the training data and evaluated on a held-out test set.

2.1 Dataset Loading and Description

The dataset used in this study contains 891 rows and 12 columns. The columns are PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, and Embarked. These columns describe passenger identity, survival outcome, ticket class, demographic details, family members aboard, fare, cabin information, and port of embarkation.

For the prediction model, the selected input features are Pclass, Sex, Age, SibSp, Parch, Fare, and Embarked. The Survived column is the target. PassengerId, Name, Ticket, and Cabin are not included in the final model inputs. Cabin also contains a large amount of missing information.



2.2 Data Quality and Missing Values

The original dataset has missing values in three columns: Age has 177 missing values, Cabin has 687, and Embarked has 2. Together, these represent 866 missing cells. Missing values need to be addressed because most machine-learning estimators cannot train directly when selected features contain missing data.

In the improved pipeline, missing numerical values are filled using the median calculated from the training data. Missing categorical values are filled using the most frequent category from the training data. This approach allows the model to process records with missing Age or Embarked values while avoiding use of test-set information during fitting.

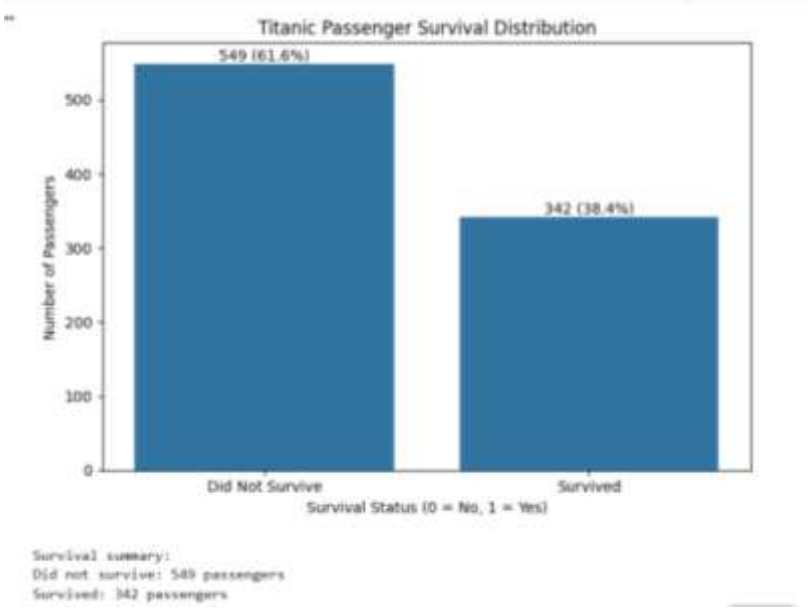
2.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is used to understand the dataset before training the model. Counts, descriptive statistics, histograms, categorical comparisons, and a correlation heatmap are examined.

Each visualization should have a clear title, labelled axes, and an interpretation that explains the observation rather than merely repeating the chart title.

2.3.1 Survival Distribution

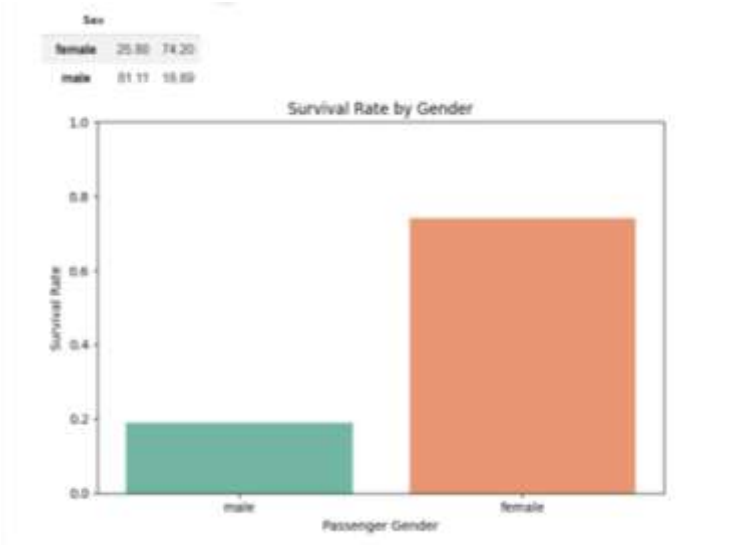
The target distribution contains 549 passengers who did not survive and 342 who survived. These correspond to approximately 61.6% and 38.4% of the dataset, respectively. The target is therefore not evenly balanced: the non-survivor class is larger. This imbalance makes it useful to consider precision, recall, F1-score, and the confusion matrix in addition to accuracy.



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2.3.2 Survival by Sex

The observed survival rate is approximately 74.20% among female passengers and 18.89% among male passengers. This is a large difference in the dataset and suggests that Sex is an informative feature for predicting survival. This finding describes an association in the historical records and should not be interpreted as a causal conclusion.

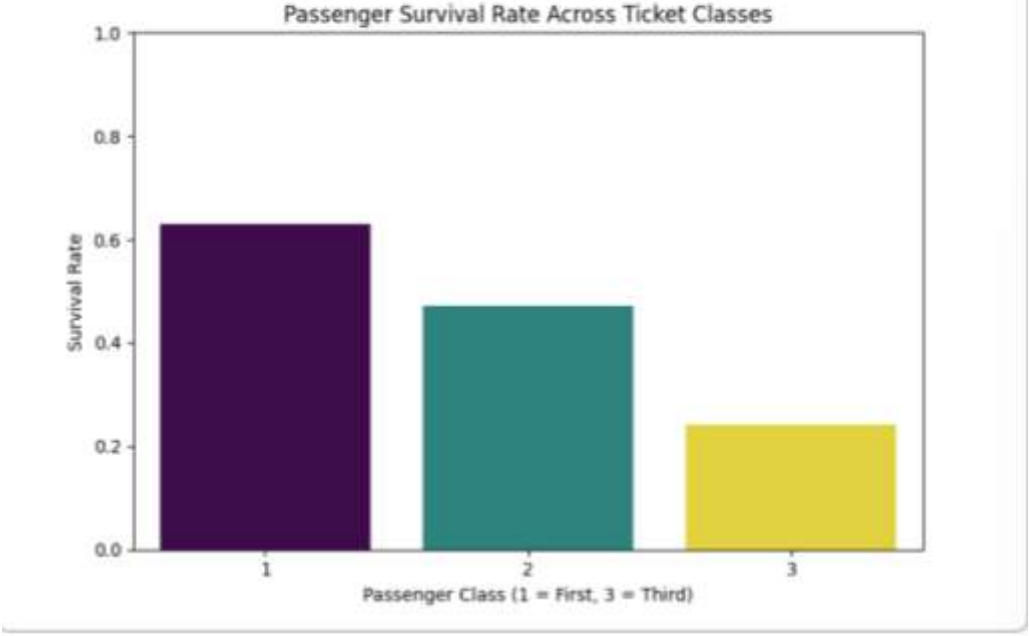


2.3.3 Survival by Passenger Class

The survival rate is approximately 62.96% for first-class passengers, 47.28% for second-class passengers, and 24.24% for third-class passengers. Survival rates decrease across these classes in this dataset. Pclass is therefore an important feature to examine in the model. The pattern may reflect differences in access, location, or other factors, but this analysis alone cannot establish the reason.

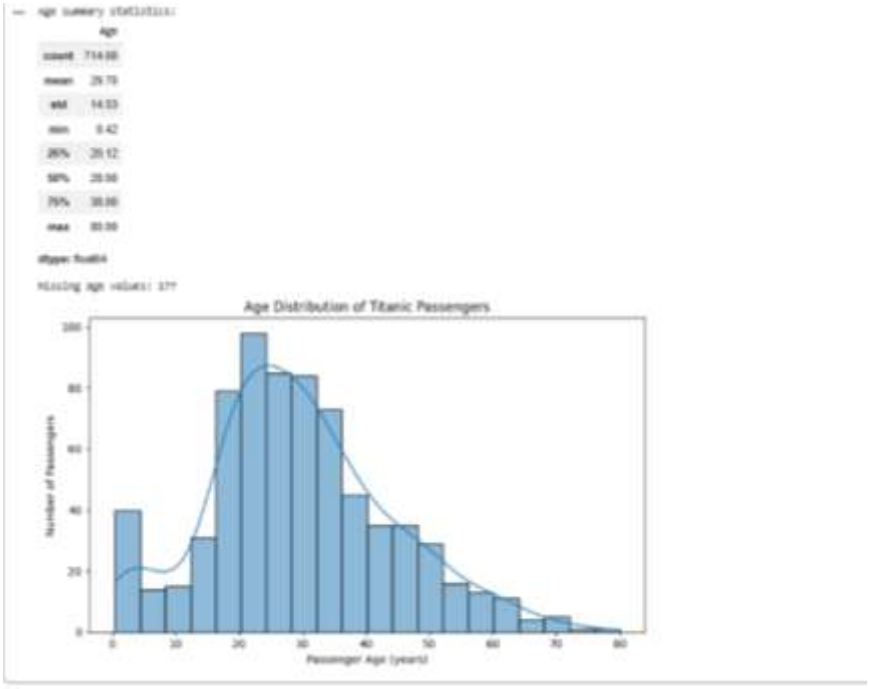
Survival statistics by passenger class:

Pclass	Passenger Count	Survival Rate	Survival Rate (%)
1	216	0.63	62.96
2	184	0.47	47.28
3	491	0.24	24.24



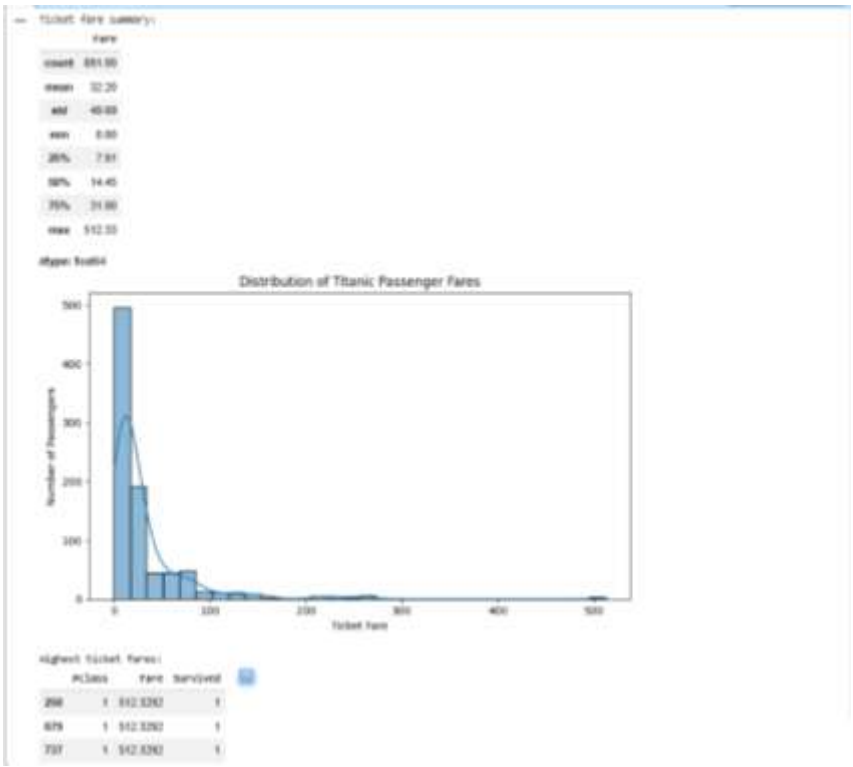
2.3.4 Age Distribution

There are 714 recorded Age values and 177 missing values. Among the recorded ages, the mean is approximately 29.70 years and the median is 28 years. The standard deviation is approximately 14.53 years, with a minimum of 0.42 and a maximum of 80. The histogram helps show the age ranges represented in the data; the missing values should be noted when interpreting this distribution.



2.3.5 Ticket Fare Distribution

The mean fare is approximately 32.20, while the median fare is approximately 14.45. The maximum observed fare is 512.33. The mean being much higher than the median and the presence of very high values indicate a right-skewed distribution. Most fares are concentrated at lower values, while a smaller number of passengers paid much higher fares.

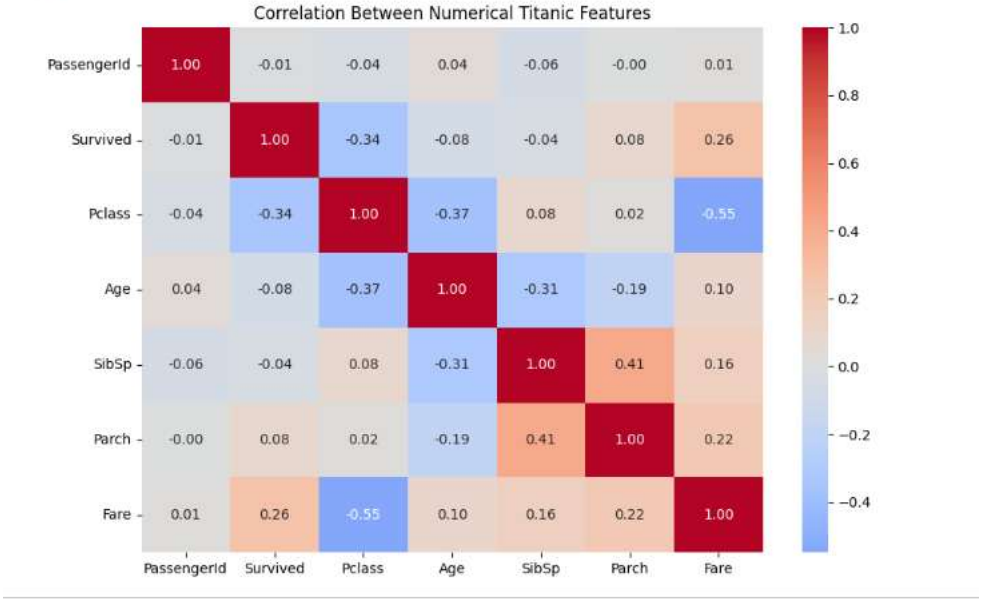


2.3.6 Correlation Analysis

The correlation between Survived and Pclass is approximately -0.338, suggesting that higher numerical class values are associated with lower survival in this dataset. Fare has a positive correlation with Survived of approximately +0.257. The correlations for Age (-0.077), SibSp (-0.035), Parch (+0.082), and PassengerId (-0.005) are comparatively weak. Correlation values describe linear association and do not prove causation. Correlations between inputs also need attention because some features may contain overlapping information.

Correlation with survival:

	Correlation with Survived
Survived	1.000
Fare	0.257
Parch	0.082
PassengerId	-0.005
SibSp	-0.035
Age	-0.077
Pclass	-0.338



2.4 Data Preprocessing

The improved modelling workflow separates the input matrix X from the target y. The input columns are Pclass, Sex, Age, SibSp, Parch, Fare, and Embarked; the target is Survived. The dataset is divided into 80% training data and 20% testing data using random_state=42 and stratification on y. This produces 712 training records and 179 testing records.

A ColumnTransformer applies different preprocessing to numerical and categorical features. Numerical features (Pclass, Age, SibSp, Parch, and Fare) use median imputation followed by StandardScaler. Categorical features (Sex and Embarked) use most-frequent imputation followed by OneHotEncoder with one category dropped as a reference. These operations are placed inside a Pipeline with Logistic Regression, so imputation, scaling, and encoding are fitted from the training data and then applied consistently to test data.

```
Missing values after preprocessing:
Survived      0
Pclass        0
Age           0
SibSp         0
Parch         0
Fare          0
Sex_male      0
Embarked_Q    0
Embarked_S    0
dtype: int64

Processed dataset shape: (891, 9)

Processed columns:
['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare', 'Sex_male', 'Embarked_Q', 'Embarked_S']
```

	Survived	Pclass	Age	SibSp	Parch	Fare	Sex_male	Embarked_Q	Embarked_S
0	0	3	22.0	1	0	7.2500	1	0	1
1	1	1	38.0	1	0	71.2833	0	0	0
2	1	3	26.0	0	0	7.9250	0	0	1
3	1	1	35.0	1	0	53.1000	0	0	1
4	0	3	35.0	0	0	8.0500	1	0	1

2.5 Model Building

Logistic Regression is a classification algorithm that estimates the probability of a binary outcome. In this study, the model estimates the probability that Survived equals 1 based on the selected passenger features. A default classification threshold is used to convert probabilities into predicted classes.

The model is fitted using the 712 training records. After training, predictions and predicted probabilities are generated for the 179 test records. The test set is held out from model fitting and is used to estimate how well the model performs on unseen records.

3. STATISTICAL FINDINGS

The exploratory analysis identifies three key observations. First, the dataset contains more non-survivors than survivors (61.6% compared with 38.4%). Second, female passengers have a much higher observed survival rate than male passengers (74.20% compared with 18.89%). Third, survival rates differ by passenger class, decreasing from 62.96% in first class to 24.24% in third class. Fare also shows a moderate positive correlation with survival, while the simple correlations for Age, SibSp, and Parch are comparatively small.

These findings guided the feature selection and provide context for evaluating the model. They should be interpreted as patterns in this dataset, not as proof that a single passenger characteristic caused survival.

4. MODEL EVALUATION

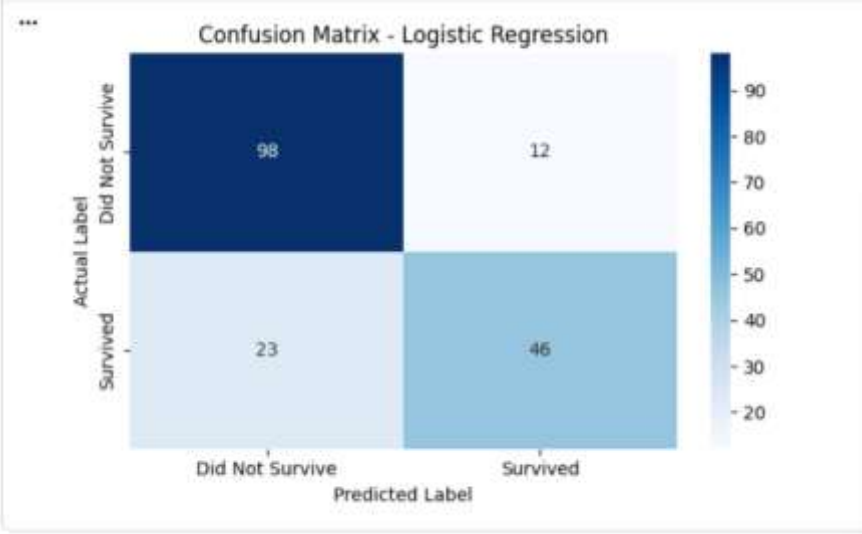
The improved Logistic Regression pipeline was evaluated on the test set of 179 passenger records. Accuracy, precision, recall, F1-score, and ROC-AUC were calculated. Multiple metrics are used because each measures a different aspect of performance.

Metric	Score	Meaning
Accuracy	80.45%	Share of all test records classified correctly.
Precision	79.31%	Of predicted survivors, the share who actually survived.
Recall	66.67%	Of actual survivors, the share identified by the model.
F1-score	72.44%	Harmonic balance of precision and recall.
ROC-AUC	84.35%	Ability to rank survivors above non-survivors across thresholds.

The accuracy of 80.45% means the model classified approximately four out of five test records correctly. Precision of 79.31% indicates that most passengers predicted to survive did survive. Recall of 66.67% shows that the model identified about two-thirds of actual survivors, so some survivors were missed. The F1-score of 72.44% summarizes the balance between precision and recall. The ROC-AUC of 84.35% indicates useful discrimination between the two classes.

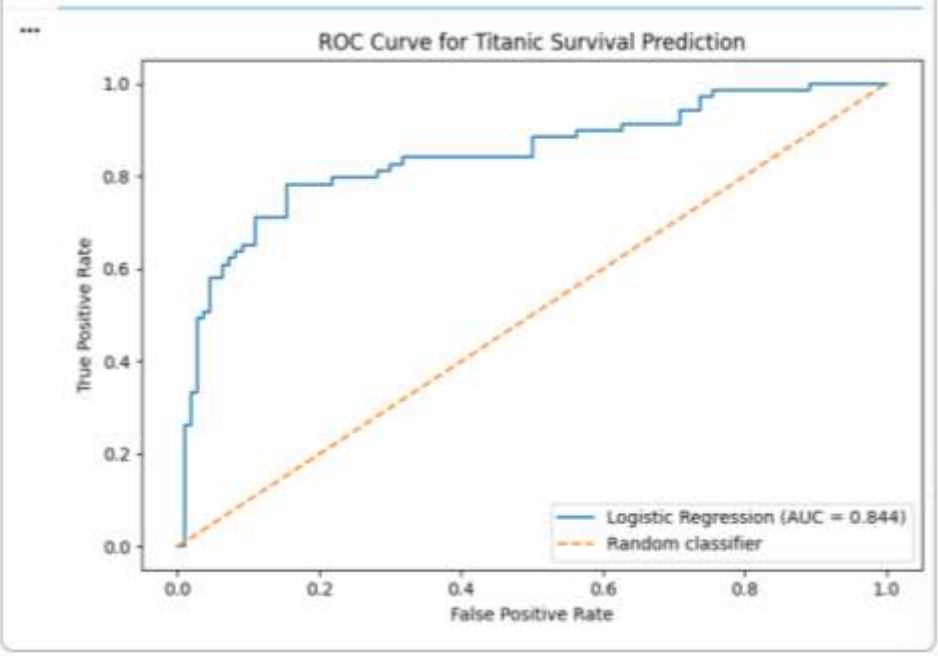
4.1 Confusion Matrix

The confusion matrix is [[98, 12], [23, 46]], with rows representing actual classes (did not survive, survived) and columns representing predicted classes in the same order. The model correctly identified 98 non-survivors (true negatives) and 46 survivors (true positives). It predicted survival for 12 passengers who did not survive (false positives) and predicted non-survival for 23 passengers who actually survived (false negatives). The model therefore correctly classified 144 of 179 test records.



4.2 ROC Curve

The ROC curve plots the true positive rate against the false positive rate at different probability thresholds. The diagonal line represents the performance of a random ranking baseline. The final improved pipeline has ROC-AUC 0.8435. For consistency, the ROC curve inserted here should be generated using y_prob from the improved pipeline; the earlier ROC plot used the original model and reported 0.8443.



4.3 Coefficient Interpretation

A Logistic Regression coefficient indicates the direction of association with the predicted log-odds of survival while the other features are held constant. Positive coefficients correspond to higher predicted log-odds; negative coefficients correspond to lower predicted log-odds. Categorical coefficients are interpreted relative to their encoding reference category.

Because the improved pipeline standardizes numerical features and one-hot encodes categorical features, the final coefficient table should be extracted from the fitted classifier inside model_pipeline and paired with the transformed feature names. Do not copy the coefficient values from the original unscaled model into the improved model's interpretation. Coefficients describe model associations and do not prove causation or statistical significance.

	Feature	Coefficient
6	Embarked_Q	0.278320
4	Fare	0.002255
1	Age	-0.038464
3	Parch	-0.071496
2	SibSp	-0.243809
7	Embarked_S	-0.381616
0	Pclass	-1.090476
5	Sex_male	-2.555755

5. CONCLUSION

This case study used Logistic Regression to predict Titanic passenger survival from passenger class, sex, age, family information, fare, and embarkation port. Exploratory analysis revealed that survival differed strongly by sex and passenger class, while the fare distribution was right-skewed and the dataset contained missing values. A preprocessing pipeline was used to impute missing values, scale numerical inputs, and encode categorical variables without fitting these transformations on the test data.

On the 179-record test set, the improved model achieved 80.45% accuracy, 79.31% precision, 66.67% recall, 72.44% F1-score, and 84.35% ROC-AUC. The confusion matrix showed 144 correct classifications, including 46 correctly identified survivors, while 23 actual survivors were missed. Overall, the model demonstrated useful classification performance, but the recall and single-split evaluation show that the result is not perfect and should be interpreted with care.

REFERENCES

1. Titanic dataset supplied and analyzed for this case study.
2. Scikit-learn documentation: Logistic Regression, Pipeline, ColumnTransformer, SimpleImputer, StandardScaler, OneHotEncoder, and classification metrics.
3. Pandas, NumPy, Matplotlib, and Seaborn documentation/course materials used for data inspection and visualization.
4. Fundamentals of Data Science course materials and faculty-provided Case Study 3 instructions.